

Milestone 2: Data Transformation Report

E-commerce Customer Behavior Analytics System

1. Executive Summary

This document details the data processing pipeline developed for Milestone 2. The primary objective was to convert raw customer behavior data into a standardized, machine-readable format suitable for exploratory analysis and future predictive modeling.

2. Data Quality & Cleaning

Cleaning Operations Performed:

- **Missing Value Treatment:** The 'Satisfaction Level' column contained null entries. We applied *Mode Imputation* to maintain the categorical distribution without introducing bias.
- **Deduplication:** A global row-wise scan identified redundant entries. Removing these ensures each customer's impact on the metrics is unique and accurate.
- **Anomaly Mitigation:** Replaced 0 values in 'Items Purchased' with 1 to prevent division-by-zero errors during calculation of spend efficiency.

3. Transformation Methodology

3.1 Encoding Categorical Data

To allow mathematical algorithms to process non-numeric data, two types of encoding were used:

Technique	Columns Affected	Description
Label Mapping	Satisfaction Level	Assigned ordinal values (0, 1, 2) to preserve the inherent rank of customer sentiment.
One-Hot Encoding	Gender, City, Membership Type	Created binary columns (0/1) for each category to handle non-ordinal geographical and demographic data.

3.2 Numerical Scaling

We utilized **Z-Score Standardization** (StandardScaler) for numerical features. This transforms the data to have a mean of 0 and a standard deviation of 1.

Why Scale? Without scaling, columns with large values (like Total Spend) would mathematically overwhelm columns with smaller values (like Age), leading to biased models.

4. Feature Engineering

Two new metrics were engineered to provide deeper behavioral insights:

- **Spend_Per_Item:** A ratio derived from `Total Spend / Items Purchased`, indicating the average price point of products favored by a customer.
- **Is_High_Spender:** A binary segmentation flag based on the 50th percentile (Median) of total spending.

5. Aggregation Insights

Final grouping by 'Is_High_Spender' revealed that high-value segments generally report higher satisfaction scores and higher average item counts, validating the data's internal consistency.